Haiyi Li

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Education

University of Adelaide, Adelaide, Australia *July 2024 - Present*
Bachelor of Mathematical Sciences (Honours) *GPA: 6.857/7.0 ranking1st*
Ocean University of China, Qingdao, China *August 2022 - July 2024*
Bachelor of Mathematics and Applied Mathematics *GPA 3.95/4.0 ranking 1st/38*
Harvard University *Upcoming Graduate Studies*
Admitted to Master's in Computational Science & Engineering

Research Interests Data-Centric AI & Scientific Computing.

Research Interests: Computational Science & Engineering; Numerical PDEs; Uncertainty Quantification; Inverse Imaging; Data-centric AI and Scientific Machine Learning.

Selected Projects

OUGS: Object-aware Uncertainty for Next-Best-View in 3D Gaussian Splatting *Nov 2024 – Present*
First Author — Eurographics 2026 (Accepted)

- Built an interpretable uncertainty propagation pipeline for 3DGS and aggregated per-Gaussian uncertainty into object-level signals for view planning.
- Designed a next-best-view strategy driven by expected uncertainty reduction; implemented the end-to-end system and performed ablations to characterize failure modes under occlusion and sparse views.
- Supported by a University of Adelaide Summer Research Scholarship (AUD 1,500).

Local LLM Pipeline for Structured Extraction from Endometriosis-Specific TVUS Reports *Mar 2025 – Present*
First Author — Submitted to CHI 2026 Posters

- Developed a privacy-preserving local LLM pipeline (86.02% accuracy) for ultrasound reports; identified complementary error patterns between AI (semantic) and humans (protocol) to design a risk-prioritized human-in-the-loop workflow.

Honours Thesis: ODE Model of Social-Structure Dynamics in Australian House Mouse *Jul 2025 – Present*
Advised by A/Prof. Edward Green — In collaboration with CSIRO (Rodent Management)

- Formulated a mechanistic ODE model coupling demographic growth with social conversion dynamics; derived equilibria and local stability conditions.

EndoExtract: Co-Designing Human-AI Interfaces for Clinical Abstraction *Mar 2025 – Present*
First Author — Submitted to CHI 2026 Posters

- Identified trust asymmetries via contextual inquiry; co-designed interface with evidence highlighting and selective review, transforming manual data entry to supervisory validation.

A Variational Path to Laplace's Equation via Complex Analysis *Jun 2025 – Present*
First Author — Submitted to American Mathematical Monthly

- Analyzed a degenerate variational functional whose Euler–Lagrange condition reduces to Cauchy–Riemann-type constraints; derived Laplace's equation as the corresponding compatibility condition.

Physiological-Sensing Awareness and AI Persuasion *Mar 2025 – Present*
Fourth Author — Submitted to DIS 2026

- Implemented repeated-measures statistical analyses (RM-ANOVA and paired nonparametric tests) with effect-size reporting for a within-subject experimental design.

Publications

Haiyi Li, Qi Chen, Denis Kalkofen, Hsiang-Ting Chen. "OUGS: Active View Selection via Object-aware Uncertainty Estimation in 3DGS." *Submitted to EuroGraphics 2026. (Accepted)* [arxiv]

Haiyi Li, Yutong Li, Yiheng Chi, Alison Deslandes, Mathew Leonardi, et al. "Who Fails Where? LLM and Human Error Patterns in Endometriosis Ultrasound Report Extraction." *Submitted to CHI 2026 Posters (accepted)* [arxiv]

Haiyi Li, Yiyang Zhao, Yutong Li, Alison Deslandes, Jodie Avery, M. Louise Hull, et al. "EndoExtract: Co-Designing Structured Text Extraction from Endometriosis Ultrasound Reports" *Submitted to Interactive Health 2026 (under review)* [arxiv]

Haiyi Li. "A Variational Path to Laplace's Equation via Complex Analysis." *Submitted to American Mathematical Monthly (under review).*

Xiaoyan Wei, Yutong Qu, Yutong Li, **Haiyi Li**, et al. "To Know or Not to Know?: How User Awareness of Physiological Sensing Impacts AI Persuasion and User Experience." *Submitted to DIS 2026. (under review)*

Awards & Honors

National Scholarship of China, *Ministry of Education of the P.R.C ($< 1\%$)* 2024
- Awarded for outstanding achievement, ranking 1st comprehensively in the cohort.

Mark Edwin Hurd Memorial Prize, *University of Adelaide (Once a year)* 2024
- Recognized for outstanding academic excellence upon entering the Mathematical Sciences honours program.

Interdisciplinary Contest in Modeling (ICM) Finalist, *COMAP ($< 2\%$)* 2024
- *Topic: Weather Changes Like the Weather Itself.* Engineered an advanced predictive model for extreme weather events, coupled with a sophisticated insurance premium estimation framework.

Summer Research Scholarship, *University of Adelaide ($< 5\%$)* 2024
- Awarded to high-potential undergraduate students in STEM majors for research excellence.

Global Citizen Scholarship, *University of Adelaide ($< 5\%$)* 2024
- Awarded for strong academic merit and contribution to the university community.

Outstanding Student, *Ocean University of China ($< 10\%$)* 2023 & 2024
- Awarded for comprehensive academic excellence and exemplary conduct.

China Mathorcup Mathematical Modeling Challenge - Second Prize ($< 10\%$) 2024
- Awarded for developing a Faster R-CNN model for efficient Oracle bone character recognition.

Technical Skills

Programming	Python, MATLAB, R, SQL	**Visualization**	Matplotlib, Tableau, Gephi
3D / Graphics	3D Gaussian Splatting (3DGS), multi-view reconstruction, NeRF	**Writing**	LaTeX, Markdown, MS Office
ML / CV	PyTorch, OpenCV, scikit-learn	**Languages**	Mandarin (native), English (proficient)
Engineering	Git, Docker, Jupyter		

Internship & Working Experience

Industrial Trainee, *CSIRO (Advisor: Dr Matthew Rees)* *Aug 2025 – Present*
- Investigated how initial class structure influences outbreak trajectories in population models, bridging industry data and dynamical systems analysis.

Research Assistant, *IMAGENDO Project, Robinson Research Institute* *Mar 2025 – Nov 2025*
- AI-assisted gynecological ultrasound: built preprocessing/data tooling and lesion-detection prototypes.

Math Tutor, *Kumon Home-based Program* *Jan 2025*
- Provided mathematical instruction and mentorship to students aged 5 to 16.

Professional Memberships

Student Member, Institute of Electrical and Electronics Engineers (IEEE) 2024 - Present

Student Member, Consortium for Mathematics and Its Applications (COMAP) 2024 - Present

Student Member, EuroGraphics (EG) 2026 - Present